

Normal Form Games · 1 topic

Game: (N, A, u) : agents, actions $A = A_1 \times \dots \times A_n$, payoffs $u_i : A \rightarrow \mathbb{R}$. 2 players + finite $\Rightarrow$ matrix (row=P1, col=P2). **Best response:** $a_i \in \text{BR}(a_{-i})$ iff $u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i}) \forall a'_i$. **Strictly dominated:** $\exists a'_i: u_i(a'_i, a_{-i}) > u_i(a_i, a_{-i})$ for every a_{-i} .

IESDS repeatedly delete strictly dominated actions; survivors = prediction.

STRICT only. Deleting **weakly** dominated can kill real NE + is order-dependent.

Pure NE profile where nobody gains by unilateral deviation. **Find:** mark each player's best-response payoff per opponent action; cell with **all** marks = NE (can be 0, 1, many).

Mixed NE — indifference method (2×2)

1. Opponent plays action w.p. p .
2. Write your EU(act₁), EU(act₂) as fns of p .
3. Set **equal** $\rightarrow$ solve p . Repeat for other player.

The p making ME indifferent = prob. the OPPONENT plays. (Nash: every finite game has ≥ 1 NE, maybe mixed.)

Worked example Rows a, b ; cols c, d ; payoffs $ac = 3, 4$, $ad = 5, 1$, $bc = 2, 2$, $bd = 8, 3$. P2 plays c w.p. p , P1 indiff: $3p + 5(1-p) = 2p + 8(1-p) \Rightarrow p = \frac{3}{4}$. P1 plays a w.p. q , P2 indiff: $4q + 2(1-q) = q + 3(1-q) \Rightarrow q = \frac{1}{4}$. Mixed NE: $(a : \frac{1}{4}, b : \frac{3}{4}), (c : \frac{3}{4}, d : \frac{1}{4})$. Each prob $\in [0, 1]$ or you slipped.

Canonical games

- **Prisoner's Dilemma:** dominance $\Rightarrow$ unique bad NE (defect, defect).
- **Matching Pennies** (zero-sum): no pure NE; mixed $(\frac{1}{2}, \frac{1}{2})$.
- **Battle of Sexes:** 2 pure NE + 1 mixed; players mix with **different** probs.

Cournot duopoly $P(Q) = a - Q$, $Q = q_1 + q_2$, cost c . $\pi_i = q_i(a - q_1 - q_2 - c)$. FOC $\Rightarrow b_i(q_j) = \frac{1}{2}(a - q_j - c)$.

Symmetric NE: $q_1^* = q_2^* = \frac{1}{3}(a - c)$.

Extensive Form Games · 2

Tree = sequential moves. Node=choice, edge=action, leaf=payoff vector. **Strategy:** **complete contingent plan:** an action at every own node / info set, even unreached ones.

Specifying unreached nodes $\Rightarrow$ normal-form redundancy + lets NE rest on **non-credible threats**.

Extensive $\rightarrow$ Normal rows/cols = full strategy sets; each cell = leaf reached by that profile; then find NE normally.

Subgame-perfect equilibrium (SPE) profile that is a NE in **every** subgame. Rules out non-credible threats. Every SPE is a NE (not vice-versa); ≥ 1 always exists. **Disprove SPE:** find one subgame where the restriction isn't a NE.

Backward induction (gives a SPE) from leaves up: at node h , mover $\rho(h)$ picks the child maximizing **their own** payoff coordinate; propagate that whole vector up.

Compare on the **moving** player's component only. Zero-sum: BI = minimax (+ alpha-beta).

Bayesian Games · 3

Game: (N, A, Θ, p, u) : types Θ_i , common prior $p(\theta)$, payoffs $u_i(a, \theta)$ depend on types.

BNE: each type θ_i best-responds in **expectation** over opponents' types:

$$\max_{a_i} \sum_{\theta_{-i}} p(\theta_{-i} | \theta_i) u_i(a_i, a_{-i}(\theta_{-i}), \theta)$$

Compute for each of your types, compute expected utility of each action over opponent types (use conditional prob.), pick best. Check no type wants to deviate. Strict dominance can be checked **per type**.

Repeated Games · 4

Discount $\beta \in (0, 1)$: weight on future; $\sum \beta^t r_t$. Higher β = more patient.

Geometric series: $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$. Constant c from now = $\frac{c}{1-\beta}$; from next period = $\beta \frac{c}{1-\beta}$.

- **Finitely** repeated: backward induction $\Rightarrow$ play stage-NE every round (cooperation unravels).
- **Infinitely** repeated: cooperation sustainable via threats.

Grim trigger (infinite PD) cooperate until a defection, then defect forever. Cooperation is a NE iff patient enough:

$$\beta \geq \frac{T-R}{T-P}$$

(T =temptation, R =reward, P =punish). i.e. future loss from punishment $\geq$ one-shot gain from defecting.

Folk thm: any feasible payoff $>$ minimax is a NE for β near 1.

Multiagent Learning · 5

Fictitious play

1. Count opponent's past actions $C_i(a)$.
2. Belief = empirical freq. $\hat{p}(a) = \frac{C_i(a)}{\sum C_i(a)}$.
3. Best-respond to belief; observe; update counts. Repeat.

If FP converges, the point is a NE. Converges for zero-sum + 2×2 games; can cycle (Shapley).

Multiagent Q-learning

- **Independent** learners: own $Q_i(s, a_i)$, ignore others (non-stationary).
- **Joint-action** learners: $Q_i(s, \vec{a})$ over joint actions $\Rightarrow$ table is $|A|^n$ vs $|A|$.
- Update: $Q(s, a) \leftarrow Q + \alpha[r + \gamma \max_{a'} Q(s', a') - Q]$. Minimax-Q / Nash-Q replace max by game value.

Auctions · 7

Second-price / Vickrey: highest bid wins, pays 2nd price. **Truthful bidding is dominant.** **First-price sealed:** highest wins, pays own bid $\Rightarrow$ **shade** your bid.

First-price, n bidders, valuations iid $U(0, 1)$, risk-neutral: symmetric eq. bid $b(v) = \frac{n-1}{n} \cdot v$.

- **English** (ascending) $\approx$ second-price; **Dutch** (descending) $\approx$ first-price.
- **VCG:** each winner pays the **externality** it imposes (others' lost welfare) $\Rightarrow$ truthful.
- **Revenue equivalence:** under iid + risk-neutral, all these auctions give the same expected revenue.

Coordination · 8

- **Social conventions:** fixed ordering over agents+actions $\Rightarrow$ each picks deterministically, no conflict.
- **Coordination graph / factor graph:** global payoff = $\sum_j f_j$ of local (often pairwise) factors.

Optimal joint action $\vec{a}^* = \arg \max_{\vec{a}} \sum_j f_j$ via **variable elimination:** eliminate agents one by one, each maximizing out over its factors (best-response fn of neighbours).

Agents & Architectures · 6,9

Agent: autonomous action in an environment to meet design objectives. Behaviours: **reactive** (timely response to dynamic env), **pro-active** (goal-directed), **social**.

Social: **coordination** = manage interdependencies; **cooperation** = shared goal; **negotiation** = agree on (conflicting) interests. Don't conflate.

Environment props accessible/inaccessible, deterministic/non-det, static/dynamic, discrete/continuous, episodic/non. (chess = "easy" side of each.) **Reactive (subsumption)** no symbolic model; behaviour layers, **lower inhibits higher**. **Limits:** local/short-term info only, **no learning**, hard to engineer many layers. **Deliberative / BDI**

Practical reasoning = deliberation (WHAT goal) + means-ends reasoning (HOW = planning).

Beliefs / Desires / Intentions. Intentions persist + filter future options. Commitment: **blind** (until achieved) / **single-minded** (until achieved or impossible) / **open-minded** (while still desired). **Hybrid layering horizontal** = all layers see I/O (needs mediator, fault-tolerant) vs **vertical** = pipeline.

MDP (safety box)

(S, A, P, R, γ) , Markov + fully observable.

Bellman opt.: $Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s' | s, a) \max_{a'} Q^*(s', a')$; $V^* = \max_a Q^*$.

Value iteration: iterate Bellman to Q^* . **Policy iteration:** evaluate + greedily improve. Markov game = MDP with joint actions + per-agent rewards $\Rightarrow$ equilibrium policies.

Misc safety one-liners

- **Rational agent** = maximizes expected utility $u^* = \arg \max_a \sum_o u(o) P(o|a)$; spec via **PEAS**.
- **Voting** (not seen on exams): Plurality / Borda ($n-1, \dots, 0$) / Condorcet winner (may not exist—cycle). **Arrow:** ≥ 3 options, Pareto+IIA $\Rightarrow$ dictatorship. **Gibbard-Satterthwaite:** non-dictatorial onto rule is manipulable.